

Code & Files (GitHub): <https://github.com/ManjuVikram/website-traffic-analysis>

Executive Summary

Analyzed 226,230 website traffic events (pageviews, previews, clicks) across music smart-link pages over one week (Aug 19–25, 2021) for Alfido Tech, covering 211 countries and 3,822 links. Total traffic declined 13.9% across the week. Overall click-through rate is a strong 39.2% of pageviews, though only 20.1% of visitors preview first, suggesting referral-driven rather than discovery-driven traffic. Traffic is geographically concentrated (Saudi Arabia + India ≈ 40%) and heavily dependent on a single track (18% of all traffic). Click-through quality varies widely by country and doesn't track with volume.

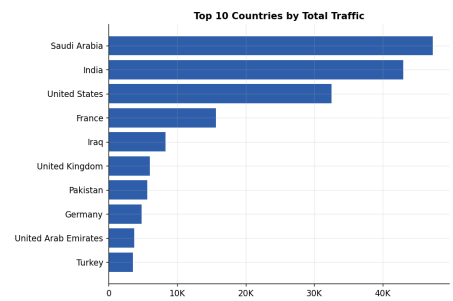
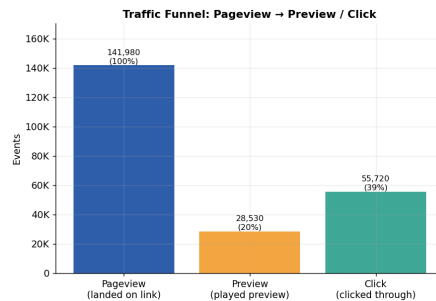
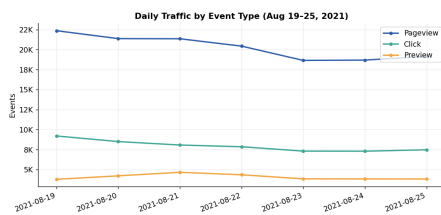
Key Findings

- Total traffic declined 13.9% over the one-week window.
- Overall CTR is a strong 39.2%, but only 20.1% preview first.
- Saudi Arabia + India $\approx$ 40% of all traffic; 2 cities dominate KSA.
- CTR doesn't track with volume — Saudi Arabia lags at 32.9%.
- One track ("Jalebi Baby") drives 18% of all traffic.

Top Recommendations

1. Investigate the cause of the 13.9% weekly traffic decline.
2. Prioritize high-CTR, smaller markets for growth investment
3. Diagnose why Saudi Arabia's CTR lags its traffic volume.
4. Reduce dependency on a single top track via cross-promotion.
5. Study and replicate what the highest-CTR top links do differently.

Key Visualizations



Daily traffic trend | Conversion funnel | Top countries by traffic

Notebook Screenshots

Website Traffic Analysis — Alfido Tech

Task 3 of 3 — Internship Data Analytics Program

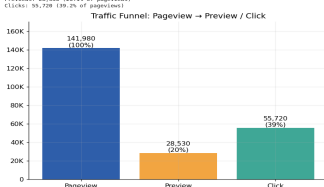
Dataset: Music smart-link traffic events — `traffic.csv`. Each row is a single traffic event (a pageview, preview, or click) recorded on a music smart-link page, with the visitor's country/city and the artist/track/album/link being viewed.

Goal: Analyze website (smart-link) traffic to understand visitor volume and trend, the pageview → click conversion funnel, geographic patterns, and top-performing content, and provide actionable recommendations for Allfido Tech.

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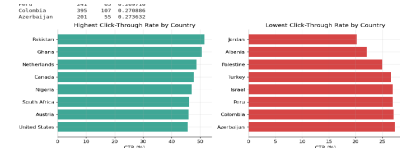
1. Setup & Data Loading
2. Data Cleaning
3. Summary Statistics & Data Types
4. Feature Engineering
5. Traffic Trend Analysis
6. Conversion Funnel Analysis
7. Geographic Analysis
8. Content Performance (Artists, Tracks, Links)
9. Key Findings & 5 Actionable Recommendations

1. Setup & Data Loading

[illegible][illegible]

Finding: 39.2% of all pageviews convert directly to a click-through — a strong headline conversion rate for this kind of link page. Only 20.1% of visitors play a preview first, meaning the large majority of clicks come from visitors who already knew what they wanted to hear, likely arriving via a direct share or promotion rather than discovering the track on the page itself.

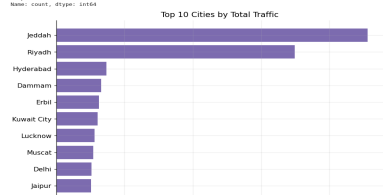
7. Geographic Analysis

[illegible]

Findings: click-through quality varies widely and doesn't track with volume — Pakistan has the highest CTR (51.5%) among high-volume countries, while Saudi Arabia, despite having the most total traffic, converts at a below-average 32.9%. Jordan, Albania, and Palestine convert lowest (20-25%), suggesting their traffic may be lower-intent or driven by a different, less targeted source.

```
In [15]: top_cities = df[df["city"] != "Unknown"]["city"].value_counts().head(10)
```

```
fig, ax = plt.subplots(figsize=(8, 5.2))
ax.set_xlabel('Year, City, and Total Values')
ax.imshow(data.index, plot_data.values, color='708080')
ax.set_title('Top 10 Cities by Total Traffic')
ax.axis('off')
ax.figure.colorbar(label='Functionality', format='%f')
plt.tight_layout()
plt.show()
```



Finding: Jeddah and Riyadh (both Saudi Arabia) are the two largest single cities by far, together exceeding the next 8 cities combined — confirming that Saudi Arabia's traffic volume is concentrated in a couple of major metro areas rather than spread nationally.

Data loading & overview | Funnel analysis with narrative findings | Geographic breakdown

Full notebook, executed outputs, and the detailed 8-page analytical report are available in the GitHub repo above.